

Customer Churn Prediction using Decision Tree

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Abstract

This paper aims to conceive and test resilient machine learning models for the classification of potential telecommunication customer churners and non-churners. From the Telco Customer Churn Dataset, we conceived and compared the accuracy of three classification models: Decision Tree Classifier, K-Nearest Neighbors (KNN) Classifier, and the high-performing ensemble method, Random Forest Classifier. The methodology included the relevant data preprocessing steps, such as missing values handling, encoding of categorical features, and scaling of numerical data for compatibility and optimal training of the model.

Introduction

The advantage of being able to predict whether a customer will churn or not is significant, especially in the Telco dataset that is used in the making of Decision Tree in RapidMiner and Python. Decision trees are algorithms implemented in machine learning that learns and generates predictions based on the attributes in a dataset. In this case, with all the attributes provided it considers all those before ending with the proper prediction. While Python utilizes libraries such as Scikit-learn which includes the appropriate modules to be used.

The main objective of customer churn prediction using decision trees is to build and evaluate a decision tree model to predict whether a customer will churn or simply, if they will cancel their subscription. The prediction will provide an advantage for Telco to build an identity for themselves as a company that values customer retention, and marketing strategies to understand their customers' preferences better.

Methodology

In predicting the customer churn of Telco, a decision tree was employed. The process, developed with the AI Studio (Rapidminer) software, and Python in Visual Studio Code (IDE) which focuses on predicting whether the customers will churn or not.

1.1 Data Collection

The first step was acquiring the raw data, and it was obtained from a file called "Telco Customer Churn Dataset." The file contained significant data that was used later on to enhance the prediction in the decision tree.

In Python, the “Telco Customer Churn Dataset” was imported from a local CSV file into the Python environment by utilizing the pandas library. This is the raw data from which all the analysis and prediction were done later on.

1.2 Data Preprocessing

After retrieving the raw dataset of Telco Customer Churn in RapidMiner, it runs through a series of preprocessing steps. With select attributes to remove the customer ID, Device Protection, Multiple Lines, Phone Service were removed as it is irrelevant in prediction using Decision Tree.

The column “gender” which contains data of “Female”, and “Male” were changed from nominal into numerical data types, where binary coding was utilized. This will contribute to a more thorough prediction.

The Python code covers a number of steps in data preparation to be used for modeling purposes. Label Encoding is first applied to all categorical data so that the text data is numerically represented, including the target variable Churn. Next, the dataset is split into features (X) and the target variable (y). This split leads to a learning set with 70% of data and a test set with 30% (obtained through varying the test_size parameter to 0.3), with stratified sampling (stratify=y). This use of stratification validates that the proportional number of churned customers remains present across both data subsets, hence overcoming the class imbalance problem.

Feature Scaling is then applied with StandardScaler to the data from the KNN model, with the effect of standardizing value across features so that larger numerical ranges do not unduly dominate distance-based algorithms.

1.3 Tools Used and Analysis Process

The data was split into 70% training set in partition 1 with and 30% testing set in partition 2. In the set role operator, the churn attribute is set as the label or target attribute to be visualized later in the decision tree. Connected to it is the Decision Tree, followed by the Apply Model operator, and later on the Performance (Classification) operator.

For comparison, in another scenario Random Forest was employed instead of Decision Tree. The Random Forest showed a slightly better accuracy in prediction, however Decision Tree was chosen as it is the better option for understanding customer behavior, specifically about whether the customer will churn or not.

In Python data analysis, there are two comparative evaluation models: a Decision Tree Classifier (DT) and a K-Nearest Neighbors (KNN) Classifier. Decision Tree relies on the 'entropy' measure and restricted max_depth=5 to maintain interpretability and prevent overfitting and trains on the baseline set of features.

The KNN model is set with n_neighbors=5 and trained on scaled data to reflect its distance-based nature. Both the models make predictions on churn according to their respective test sets, then their performances are thoroughly validated through a common classification set of metrics: Confusion Matrix, Accuracy, F1 Score, Precision, and Recall.

Results

accuracy: 79.03%

	true No	true Yes	class precision
pred. No	1373	247	84.75%
pred. Yes	196	297	60.24%
class recall	87.51%	54.60%	

In RapidMiner the Decision Tree achieved an accuracy of 79.03% which gives us the conclusion that the model is more accurate at identifying the customers who will stay ("No") than those who will churn ("Yes").

f_measure: 57.28% (positive class: Yes)

	true No	true Yes	class precision
pred. No	1373	247	84.75%
pred. Yes	196	297	60.24%
class recall	87.51%	54.60%	

The F1-score for the churn class ("Yes") is 57.28%, this indicates that while the model balances precision and recall well. It struggles to identify all customers who will churn correctly.



In the decision tree, to summarize the findings the most significant predictors were the Tenure, Monthly charges, and Total charges. Customers who subscribed to the company for more than 5.5 months have a lower chance of churning, while customers that subscribed for less than or equal to 5.5 months have a higher chance of churning, most especially when the total charges are low. With monthly charges above 93 also adds as a factor in churning, as well as senior citizens with certain billing or charges. However, if a customer is subscribed to the company for a long time and has high total charges of greater than 8,193 (as shown in the split) are likely to remain subscribed. To simplify, new customers or those who have high monthly charges are most likely to churn, while long-term customers with higher total charges are loyal and will remain subscribed.

accuracy: 80.41%

	true No	true Yes	class precision
pred. No	1456	301	82.87%
pred. Yes	113	243	68.26%
class recall	92.80%	44.67%	

To compare, we used Random Forest which has a slightly improved accuracy of 80.45% than Decision Tree with 79.03%. While it does not provide similar clear rules, it confirms that tenure, charges, and the status of senior citizen remains in the most influential features.

Model Performance Comparison

Decision Tree
Accuracy: 78.70%

K-Nearest Neighbors
Accuracy: 74.35%

	true No	true Yes		true No	true Yes
pred No	1331	221	pred No	1289	263
pred Yes	229	332	pred Yes	279	282
class precision	85.32%	60.04%	class precision	82.21%	51.74%
class recall	85.76%	59.18%	class recall	83.05%	50.27%

In Python, a 70% training and 30% testing split was utilized to compare the performance of the Decision Tree (DT) and K-Nearest Neighbors (KNN) models.

The Decision Tree model showed a higher of 78.70% for the whole accuracy compared to 74.35% for KNN model. Both models, however, showed considerably higher accuracy for classification of the non-churn category ("No") due to the dataset imbalance, manifested in high Non-Churn Precision (DT: 85.32%, KNN: 82.21%) and Recall (DT: 85.76%, KNN: 83.05%).

As for the important Churn ("Yes") class, the best model was the Decision Tree. Its Precision (60.04%) means 60.04% of its predicted churners are accurate, and its Recall (59.18%) means it correctly identified 59.18% of all actual churners. Compared to the KNN model, it performed far less well, especially in Recall (50.27%), missing almost half of the actual churning customers. The latter's higher quantity of False Negatives over DT (279 vs. 229) again proves the superiority of the Decision Tree to capture the difficult minority class.

Discussion

1.1 Interpretation of Results

From the results of the Decision Tree in RapidMiner, it suggests that the company should pay more attention to the new customers during their first months of subscription and as well as the subscribers that have higher monthly charges, since they are the most at risk of leaving or unsubscribing. To reduce the churn, strategies such as loyalty programs, discounts, or enhanced experiences may help to lower the possibility of them churning. Meanwhile, the loyalty of long-term customers with high total charges highlights the value of maintaining strong customer relationships over time.

As for the analysis using Python, the effectiveness of the Decision Tree model in optimizing lost revenue is based on its enhanced Recall of the churn class. The foremost and first requirement is utilizing the model able to identify the maximum number of true churners, therefore garnering a dominant advantage for early intervention. The Churn Recall of the Decision Tree of 59.18% dramatically exceeds the K-Nearest Neighbors model with Churn Recall of 50.27%. This indicates the Decision Tree is able to correctly identify nearly sixty percent of all the clients who are on the brink of abandoning the organization. By deploying the DT model, the company immediately gains the necessary lead time for proactive customer relationship management (CRM).

1.2 Limitations

The RapidMiner's Decision tree accuracy reached 79.03% with an F1-score of 57.28% for churn, this means that it is much better at predicting customers who will stay than those who will be leaving. The dataset is imbalanced with more customers staying than leaving. This affects the model's ability to capture churn cases. In connection with the imbalanced dataset, it also doesn't have other features such as satisfaction scores that could improve the predictions. Lastly, the results may not generalize well to different companies or industries, due to the dataset being specific to telco context.

As for Python, despite the use of stratified sampling so that there is an equal representation of the imbalance in the train/test sets, there is still a challenge for the models to achieve a high accuracy on the minority class. This liability becomes evident when seeing the high precision for the "No" class (85.32% for DT) compared to low precision for the "Yes" class (60.04% for DT). The low recall means there are still a

number of the future churners being missed as False Negatives. The KNN model perpetuates this liability with a still low recall of 50.27%.

1.3 Implications

The findings of this customer churn prediction from RapidMiner and Python can be a helpful asset for the Telco in order to retain their loyal customers, and as well as gain trust and loyalty from their new customer. Furthermore, they can employ this in their marketing strategy to understand their customers better and create an identity for themselves as a company that values customer retention. By identifying the new customers and those with high monthly charges as more likely to churn, the company can design targeted promotions, loyalty programs, and personalized support that can reduce churn and improve overall service quality.

Conclusion

Using the Telco Customer Churn dataset, a Decision Tree model was built using RapidMiner to predict whether the customers are likely to churn or not. The process involved preprocessing and selecting the relevant attributes to improve model performance. The Decision Tree achieved an accuracy of 79.03%, this indicates that the machine operates well at predicting customers who will stay but is less effective at identifying those who will churn. This was also tested using a Random Forest model that achieved a slightly higher accuracy of 80.45%. However, Random Forest is more complex and less interpretable which makes it harder to visualize the decision rules. Overall, both highlight significant predictors such as tenure, monthly charges, and total charges that provide valuable insights into customer churn behavior.

Meanwhile, the Python analysis confirms the best-performing model from among those tested to be the Decision Tree due to its combination of predictive accuracy, especially for the minority churn class, and intrinsic interpretability to achieve the goal of discovering rules for customer churn. The persistent low recall for the churn class for all of the tested models means there remains a point for future work to mitigate the Telco dataset's inherently imbalanced nature using further techniques.

References

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